

Challenge (Habanero)

- Q1.** A hiker's backpack weighs 2^3 pounds. How many pounds does it weigh?
(A) 4
(B) 6
(C) 8
(D) 10
(E) 12
- Q2.** A wildlife reserve has 3^2 acres of land for animal habitats. If they add 2^2 more acres, what is the total number of acres?
(A) 5
(B) 7
(C) 9
(D) 11
(E) 13
- Q3.** The population of a certain animal species is 4^2 . If the population doubles, what is the new population?
(A) 16
(B) 32
(C) 64
(D) 128
(E) 256
- Q4.** A car's fuel consumption is 2^2 miles per gallon. If the car travels 3^2 miles, how many gallons of fuel are consumed?
(A) 4.5
(B) 5.5
(C) 6.5
(D) 7.5
- (E) 9
- Q5.** Simplify the expression $(2^2)^3$.
(A) 2^5
(B) 2^6
(C) 2^7
(D) 2^8
(E) 2^9
- Q6.** Simplify the expression $\frac{2^4}{2^2}$.
(A) 2
(B) 2^2
(C) 2^3
(D) 2^4
(E) 2^5
- Q7.** Simplify the expression $2^2 \cdot 2^3$.
(A) 2^4
(B) 2^5
(C) 2^6
(D) 2^7
(E) 2^8
- Q8.** The area of a square hiking trail is 3^4 square feet. What is the length of each side of the trail?
(A) 3^2
(B) 3^3
(C) 3^4
(D) 3^5
(E) 3^6

Open Ended Questions (FRQ)

- Q9.** A national park has x acres of land, where $x = 2^5$. If $\frac{1}{4}$ of the land is used for hiking trails, what is the area of the hiking trails in terms of x ? Show your work and simplify your answer.
- Q10.** The population of a certain animal species is modeled by the equation $P = 2^t$, where t is the number of years. If the population doubles in 3 years, what is the value of t when the population is 4 times its initial value? Show your work and simplify your answer.

Chef's Tips

- Read each question carefully before answering
- Use the correct order of operations when simplifying expressions
- Check your work for reasonableness and accuracy